Serendib University Endowment Fund
Investment Policy Recommendation

A defensive USD 10 million mandate,
and the evidence for how badly it could go wrong

Team DOTTS

How should a conservative endowment split USD 10 million across seven ETFs, and how badly could that portfolio go wrong?

Summary of findings. We recommend that the committee **accept the mandate with a 20% maximum drawdown tolerance**. The portfolio holds 75% in bonds and gold and 25% in equities and meets every limit we set. In the 2008 crisis its deepest fall was −18.1%, against −30.3% for a classic 60/40 fund over the same window. Its risk model, tested out of sample from 2007, has the right number of breaches and is the only one of nine that passes the joint test of breach count and timing, though only just. Two risks are live today: the stock and bond relationship the portfolio relies on broke in 2020, a break we date statistically, and model breaches still tend to bunch, as they have over the last year. Each has a pre-agreed response rather than a footnote.

Sample: 2004-11-19 to 2026-10-26 (5,517 daily returns, 21.9 years).
Out of sample backtest: 4,985 daily forecasts from 2007-01-03.
Submission deadline: 31 October 2026.

1. Executive summary

The decision. Accept the mandate with a 20% maximum drawdown tolerance. Invest USD 10 million as **AGG 55%, GLD 20%, SPY 15%, XLE 5%, XLK 5%**, rebalanced quarterly, and pre-approve one action: lifting gold to 25% if the stock and bond correlation stays high.

What the client can expect	**How bad it could get**
Return 7.3% a year, 5.5% above T-bills	Worst fall since 2004: −18.7%
Volatility 7.3%, Sortino ratio 1.07	Average loss, worst 1% of days: $181,583
3.5% chance of a 20% fall in five years	Deepest crisis fall: −18.1% (2008)
17.7% chance of missing the return target	Worst invented crisis: −16.8% ($1,675,000)

The client told us that avoiding big losses matters more than chasing big gains. We took that literally: we chose the portfolio whose worst case we could measure, test and defend, not the one with the best historical return. Six findings shape the recommendation.

1. A 75% defensive core meets every limit at a small cost in return. The mix returned 7.3% a year against 8.1% for a 60/40 fund, but fell −18.7% at worst against −34.7%. All 9 mandate checks pass on the full sample, and all four that can be measured on recent data pass on 2021–26; on 2020–26 data alone, volatility and ES breach because of March 2020. (Sections 3 and 10)

2. Its tail is close to the best achievable for its return. Among all mixes inside our limits that earn our return, the lowest 99% ES is 1.76%; ours is 1.82%. That mix holds no XLE and would have lost −16.5% in 2022, against our −12.8%. (Section 3)

3. A 15% drawdown limit rules out meaningful equity. Within our own AGG cap, no mix in a grid of 1,601 stays inside 15% (the best falls −15.2%); even with the cap lifted, the only mixes that do hold about 5% in equities and earn 5.2%. 20% is about five years of a 4% spending payout. (Sections 2 and 3)

4. Risk numbers disagree because the tails are fat. Daily returns have excess kurtosis of 19.4. A bell curve puts the 99% one day ES at $119,779, 34% below the $181,583 that history shows. We hold the largest of three fat tailed estimates against the limit. (Sections 4 and 5)

5. The risk model passes, including 2008. Tested on 4,985 unseen days from 2007, it breached its 99% VaR 59 times against 50 expected (Kupiec $p = 0.21$) and never entered the Basel red zone. It is the only model of nine that passes conditional coverage, marginally ($p = 0.052$); its breaches still show some clustering (independence $p = 0.038$). (Section 6)

6. The bond hedge broke in 2020. A formal break test dates a change in how AGG moves with SPY to 26 March 2020 ($F = 86$ against a 5% critical value of 8.68), with a second step on 29 March 2022, as the Fed began raising rates into inflation. Correlation went from −0.08 to 0.20. This is the portfolio's main structural risk. (Section 8)

A portfolio we rejected. Our first draft spread the money more evenly: AGG 45%, SPY 20%, GLD 15%, EFA 10%, EEM 4%, XLE 3%, XLK 3%. It looked diversified and earned 7.5% a year, but failed three of our four limits: volatility 9.2%, 99% ES 2.4% and a drawdown of −25.9%. In 2008 it lost −15.3% against −4.2% for the final mix. More funds did not mean more diversification, because EFA moves almost one for one with SPY (daily correlation 0.88).

2. The mandate and the data

2.1 The mandate we propose

The brief fixes the client, the capital and the seven funds, and leaves the horizon, return target and risk limits to us. We anchor them on one assumption about the client: a spending rule that pays out 4% of the fund a year, a common endowment convention. Table 1 records each choice.

The app lets the committee move any limit and see the verdict change.

Table 1: Mandate decisions. Every value is a team decision held in `src/config.py`. Source: Team analysis.

Item	Our choice	Why
Horizon	5 years, monitored daily, reviewed quarterly	Long enough to ride out a crisis and its recovery (the 2008 drawdown took 214 trading days to recover), short enough to judge against.
Return target	T-bills + 2.5% a year	Moves with interest rates. At today's 4.0% bill rate it is 6.5%: enough to fund a 4% payout and keep up with about 2.5% inflation. If bill rates fall well below 4% it no longer covers both and should be revisited.
Volatility ceiling	8% a year	About 70% of a 60/40 fund's 11.5%.
99% one day ES	$\leq$ 2% of capital ($200,000)	About two thirds of a 60/40 fund's 3.1%. ES, not VaR, because the client fears how deep losses go.
Maximum drawdown	$\leq$ 20%	About five years of a 4% payout. We first tried 15%, which rules out meaningful equity (Section 3.3); it stays as a review trigger.
Stress loss	$\leq$ 25%	No replayed or invented crisis may cost more than a quarter of the fund, about six years of payout.
Allocation	Long only; AGG 35% to 60%; equity $\leq$ 60%; GLD $\leq$ 20% (25% if trigger (a) fires); EEM, XLE, XLK $\leq$ 5% each	A floor in a bond fund of thousands of issues, a cap on a single commodity, and small volatile satellites so that none can dominate risk.

2.2 Data, and every cleaning step

Sources. Daily adjusted closes from Yahoo Finance, so dividends and splits do not distort returns, and the 3 month Treasury bill rate (DTB3) from FRED as the risk free rate. The sample starts on GLD's first trading day. A frozen snapshot in `data/` reproduces every number with no network.

Log returns are used for all statistics; simple returns wherever funds are combined, because only simple returns add up across weights.

Calendar and cleaning. All seven funds trade on US exchanges and share one calendar. The code in `src/data.py` removes duplicate dates, forward fills gaps of up to two days (a zero return) and drops anything still missing; it flags non-positive prices, stale runs of five or more identical closes, and moves more than ten robust standard deviations from the median, measured with the scaled median absolute deviation so the crashes do not inflate the yardstick. It flagged 55 moves, 50 of them in late 2008 or March 2020. They are genuine crisis days, so we flag them and keep them: a tail risk model that deletes its tail is not worth building.

In Table 2, 6 of seven funds are negatively skewed and excess kurtosis runs from 6.6 (gold) to 55.3 (AGG), so no fund is close to a bell curve. AGG's kurtosis comes from bond ETF price dislocations in October 2008 and March 2020: on 2008-10-10, the portfolio's worst day, AGG alone fell –6.8%. The safest fund on average is not safe on every day.

3. Layer 1: how should we split the money?

3.1 Eight candidates and the trigger mix, scored on the client's terms

We built eight portfolios on the same data, plus the trigger mix of Section 8 ("Gold 25"), and scored each on what the committee cares about (Table 3). Markowitz optimisation is the starting point, but left unconstrained it gives an answer no conservative client should hold: the long only maximum Sharpe portfolio puts AGG 36%, XLK 34%, GLD 30% to work, a bet on one sector and

Table 2: Daily log return diagnostics. Jarque–Bera rejects normality for every fund; ADF rejects a unit root for every fund, so returns are stationary. ADF lags are chosen per fund by AIC, so only the test decision, not the statistic, is comparable across funds. Source: Team analysis.

Fund	Ann. vol	Skew	Ex. kurt.	Worst day	JB p	ADF stat.	ADF p
SPY	18.8%	−0.30	14.8	−11.6%	<0.001	−16.7	<0.001
EFA	20.9%	−0.35	13.7	−11.8%	<0.001	−16.2	<0.001
EEM	27.5%	0.02	15.6	−17.6%	<0.001	−16.2	<0.001
AGG	5.2%	−2.09	55.3	−7.1%	<0.001	−15.8	<0.001
GLD	18.3%	−0.45	6.6	−10.8%	<0.001	−75.4	<0.001
XLE	29.8%	−0.71	12.9	−22.5%	<0.001	−18.2	<0.001
XLK	22.6%	−0.16	9.4	−14.9%	<0.001	−18.5	<0.001

one commodity that falls −23.5%.

Table 3: The recommended mix against seven alternatives and the pre-agreed trigger mix, full sample. Weights in percent. ES is the historical 99% one day Expected Shortfall. Crisis replays hold the starting weights through each window. "C. Mkz" is maximum Sharpe and "Min ES" is minimum CVaR (Rockafellar and Uryasev, 2000), both under the mandate's allocation limits. Gold 25 meets the limits only under the trigger's 25% gold cap. Source: Team analysis.

Fund / metric	**Ours**	Gold 25	Draft	C. Mkz	Min ES	Max SR	Min var	1/N	60/40
SPY	15	15	20	24	19	–	6	14	60
EFA	–	–	10	–	–	–	–	14	–
EEM	–	–	4	–	–	–	–	14	–
AGG	55	50	45	51	60	36	92	14	40
GLD	20	25	15	20	20	30	1	14	–
XLE	5	5	3	–	–	–	1	14	–
XLK	5	5	3	5	1	34	–	14	–
CAGR	7.3%	7.7%	7.5%	7.5%	6.5%	10.2%	3.7%	10.0%	8.1%
Excess over T-bills	5.5%	5.9%	5.7%	5.8%	4.8%	8.5%	2.0%	8.3%	6.4%
Volatility	7.3%	7.8%	9.2%	7.6%	6.6%	10.1%	5.0%	15.8%	11.5%
Sortino	1.07	1.08	0.92	1.09	1.03	1.21	0.55	0.81	0.83
Max drawdown	−18.7%	−19.3%	−25.9%	−19.4%	−16.5%	−23.5%	−16.8%	−42.4%	−34.7%
99% 1-day ES	1.8%	1.9%	2.4%	1.9%	1.6%	2.4%	1.3%	4.2%	3.1%
2008 replay	−4.2%	−3.3%	−15.3%	−6.9%	−1.7%	−7.5%	3.3%	−34.1%	−30.0%
2020 replay	−10.7%	−10.7%	−14.8%	−10.8%	−8.1%	−11.8%	−3.8%	−26.6%	−20.6%
2022 replay	−12.8%	−12.5%	−16.3%	−16.8%	−15.6%	−19.1%	−14.8%	−12.5%	−20.4%
All limits met	Yes	Yes	**No**	Yes	Yes	**No**	**No**	**No**	**No**

Two optimisers obey our limits and deserve a direct answer.

Constrained Markowitz (AGG 51%, SPY 24%, GLD 20%, XLK 5%) beats us narrowly on return (7.5% against 7.3%) and Sortino (1.09 against 1.07), passes every limit and is 8.7% turnover away. We still prefer ours. It lost −16.8% in the 2022 rate shock against our −12.8%, and it rests on full sample average returns, the least reliable input in finance: Section 3.4 shows that weights chosen this way do not hold up out of sample.

Minimum CVaR optimises the measure our mandate is written in, the average loss on the worst 1% of days, directly over the historical scenarios with no assumption of normality. It holds AGG 60%, GLD 20%, SPY 19%, XLK 1%: a lower ES (1.6%) and drawdown (−16.5%), but 6.5% a year instead of 7.3% and a worse 2022 (−15.6%). Adding a return floor equal to ours to the same programme gives the lowest ES any allowed mix can have at our return: AGG 54%, SPY 21%, GLD 20%, XLK 5%, with an ES of 1.76% against our 1.82%. It holds no XLE and would have lost −16.5% in 2022, against our −12.8%. All three methods put AGG and gold at or near their caps, which the caps partly build in; the agreement is reassuring, not independent proof.

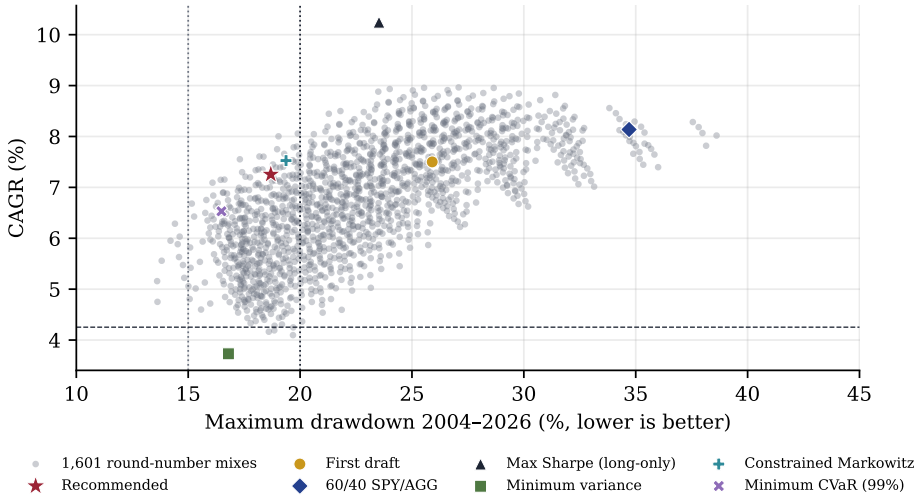


Figure 1: Return against maximum drawdown for 1,601 round number mixes and the named candidates. The grid lets AGG run to 80%, above our 60% cap, to find the true drawdown floor. The dotted lines mark 15% and 20% drawdown, the dashed line the return target. Source: Team analysis.

3.2 Why each fund earns its place

AGG (55%) is the core: in 2008 it returned 7.9% while SPY lost –36.8%. **GLD (20%)** covers the case where bonds fail too: in 2022 AGG lost –13.0% and gold –0.8%. It is capped because one commodity can go years without gaining. **SPY (15%)** and **XLK (5%)** provide the growth that pays for scholarships. **XLE (5%)** is a regime hedge, not a crash hedge. On the portfolio's worst 1% of days it fell –5.5% on average, more than SPY (–3.8%), and it lost –39.0% in 2008 and –18.2% in 2018. It earns its place only when inflation drives the market: it rose 64.3% in 2022, when stocks and bonds fell together. Without it (its 5% in AGG instead) the mix would have had a lower ES (1.61%) and drawdown (–16.8%) but returned 6.7% a year and lost –16.0% in 2022. That is why it is held, and why at only 5%. **EFA and EEM are left out.** EFA's correlation with SPY is 0.88 daily and 0.86 monthly, so it adds currency risk without diversification; replacing 5% of SPY with EFA lowers the Sortino ratio in both halves of the sample (1.18 against 1.21 in 2004–2011, 0.93 against 1.00 in 2012–2026). EEM helps only in the 2004 to 2011 emerging market boom (1.23 against 1.21, then 0.90 against 1.00).

3.3 Why the drawdown tolerance is 20%

Our mix (Figure 2) fell –18.7% between 2008-05-20 and 2008-10-27 and recovered by 2009-09-02. We first tried a 15% tolerance and tested it before deciding. Within our own 60% AGG cap, no mix in the grid (Figure 1) meets it: the best falls –15.2%. Lifting the AGG cap does not help a portfolio that holds equities: with at least 15% in equities the best drawdown is –15.8% even with AGG uncapped. The only mixes inside 15% hold 75% AGG and 5% equity and earn 5.2%, below today's 6.5% target. Only 31% of all mixes stay inside 20%, so 20% is still a demanding limit. We keep 15% as a trigger for a formal review.

3.4 Is the choice robust, or just fitted to the past?

Our weights were chosen after seeing 2004 to 2026, so their full sample record flatters them. Three checks limit that bias.

Walk forward. At each split date we picked the best Sortino mix that met the risk limits (volatility, ES and drawdown, not the allocation caps) using only earlier data, then scored it afterwards. Chosen before 2012 it fell from a Sortino of 1.45 to 0.49; chosen before 2016, from 1.15 to 0.76. Our mix scored 1.00 and 1.11 over the same later periods. Chasing the best past mix is unstable, which is why our weights are round numbers chosen by judgement.

Sub periods. Our mix held up in both halves of the sample: a Sortino of 1.21 in 2004–2011

and 1.00 in 2012–2026. The 60/40 fund beat us on Sortino in the second half, a long calm bull market, after falling –34.7% in the first. That is the trade we recommend: less in the good years, much less pain in the bad ones.

Five year bootstrap. We simulated 3,000 five year paths from one month blocks of real history, which keeps volatility clustering inside each block (Künsch, 1989). Our mix has a 3.5% chance of falling more than 20% within five years, against 33.0% for 60/40, a 1.4% chance of losing money, and a 17.7% chance of missing the return target.

Rebalancing. Our figures assume daily rebalancing; Section 10 gives the quarterly figures, which stay inside every limit.

3.5 Scorecard

Against T-bills the mix has a Sharpe ratio of 0.75 and a Sortino ratio of 1.07 (60/40: 0.59 and 0.83), a Calmar ratio of 0.39 and an Omega ratio of 1.20. Beta and alpha come from the Capital Asset Pricing Model (Sharpe, 1964), estimated by least squares of the portfolio's daily excess return on SPY's. A beta of 0.27 means most of the return does not come from the stock market. The alpha of 2.7% a year (t = 2.40) is diversification, not skill: 66% of it comes from gold's strong run over this sample, which we do not assume will repeat. The information ratio against 60/40 is –0.16 because we earn less in rising markets by design.

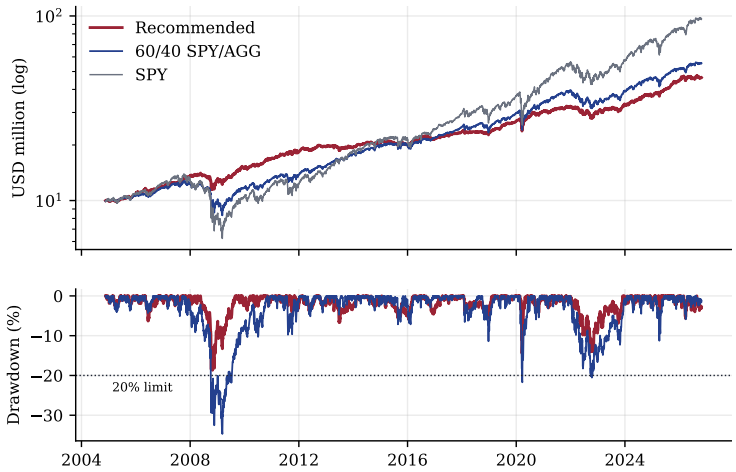


Figure 2: Growth of USD 10 million (log scale, top) and drawdown from the previous peak (bottom). The recommended mix stayed inside the 20% line through 2008, 2020 and 2022. Source: Team analysis.

4. Layers 2 and 3: how much could we lose on a bad day?

4.1 Nine estimates of the same number

Value at Risk (VaR) is the loss we should not exceed on 99 days in 100. Expected Shortfall (ES) is the average loss on the other day. We limit ES, not VaR, because ES says how deep the bad zone goes and, unlike VaR, never penalises diversification (Artzner et al., 1999). We estimated both nine ways (Table 4): the brief's historical, parametric and Monte Carlo methods, plus fat tailed and conditional versions, because the disagreement between them is itself evidence. Appendix C lists every model setting.

4.2 Why they disagree

The bell curve gets the shape wrong both ways. At 95% the Normal VaR ($73,037) is *higher* than history's ($68,021). At 99% it is lower ($104,280 against $121,735) and its ES is 34% below history. Fat tails mean more quiet days and more violent days than a bell curve allows, so the Normal model over warns on ordinary bad days and under warns on the ones that matter. The fitted Student-t has only 3.5 degrees of freedom.

Table 4: One day VaR and ES for the recommended mix, USD on USD 10 million, full sample. Monte Carlo uses 50,000 draws from the funds' joint distribution. Source: Team analysis.

Method (1-day, USD 10m)	VaR 95%	ES 95%	VaR 99%	ES 99%	ES 99% (% cap.)
Historical simulation	\$68,021	\$105,670	\$121,735	\$181,583	1.82%
Parametric (Normal)	\$73,037	\$92,205	\$104,280	\$119,779	1.20%
Parametric (Student-t)	\$62,051	\$99,223	\$116,899	\$171,853	1.72%
Cornish-Fisher	\$61,081	\$230,420	\$321,362	\$557,874	5.58%
Monte Carlo (Normal)	\$73,620	\$92,354	\$103,905	\$118,863	1.19%
Monte Carlo (Student-t)	\$64,814	\$102,497	\$121,723	\$175,433	1.75%
Filtered HS (EWMA)	\$67,261	\$99,138	\$111,720	\$157,052	1.57%
EVT (GPD-POT)	\$66,346	\$106,105	\$127,352	\$179,330	1.79%
GARCH-EVT (conditional)	\$61,444	\$90,285	\$106,973	\$139,652	1.40%

Monte Carlo is only as good as what it simulates. Normal Monte Carlo (\$118,863) almost exactly matches the Normal formula (\$119,779). Simulating a Student-t instead gives \$175,433, close to history. Simulation adds flexibility, not truth.

Conditional models read today's market. Filtered historical simulation (FHS; Barone-Adesi et al., 1999) rescales history to today's EWMA volatility (RiskMetrics, 1996), currently 0.89 times its long run average, so it reports a lower ES today (\$157,052) and rises within days in a crisis. That is why it passes the backtest in Section 6.

The number we hold against the limit is the largest of FHS, EVT and historical simulation: today that is history, \$181,583 or 1.82% of capital. A bootstrap gives a 90% interval for it of \$157,573 to \$205,090; the upper end is above the limit, so the limit is met on our best estimate, not with certainty.

Beyond one day. Scaling the log loss by the square root of ten gives a ten day ES of \$563,023; because losses cluster (Section 6) that may understate a crisis. A Basel style stressed ES, using the 2008 window only, is \$322,842 for one day. Neither has a limit of its own: the ES limit is a daily monitoring tool, and multi-day crises are governed by the 25% stress limit. Both are reported alongside the headline.

> **A result we discarded.** Cornish–Fisher, often recommended for fat tails, reports a 99% ES of \$557,874: three times every other estimate, because its polynomial stops behaving like a distribution at kurtosis of 19.4. In the backtest it breached only 39 times against 50, and those breaches clustered. It stays in the app as a warning and in no limit.

5. Layer 4: what about once in a decade disasters?

5.1 Method

Extreme Value Theory (EVT) models only the tail. We choose a loss threshold u, take the losses beyond it, and fit a Generalised Pareto distribution with shape ξ and scale β by maximum likelihood. For a high enough threshold the Pickands–Balkema–de Haan theorem says those excesses are Generalised Pareto whatever ordinary days look like, and VaR and ES follow in closed form (McNeil, Frey and Embrechts, 2015), even where history has only a handful of observations.

5.2 How we chose the threshold

The threshold is the one judgement EVT cannot make for itself: too low and ordinary days bias the tail, too high and too few observations remain. We made the usual visual rule reproducible. We fit the tail at every threshold from the 80th to the 98.5th percentile and choose the lowest one, at or above the 85th, whose shape estimate is within one standard error of the median shape of all higher thresholds. That is where the stability plot flattens (Figure 3).

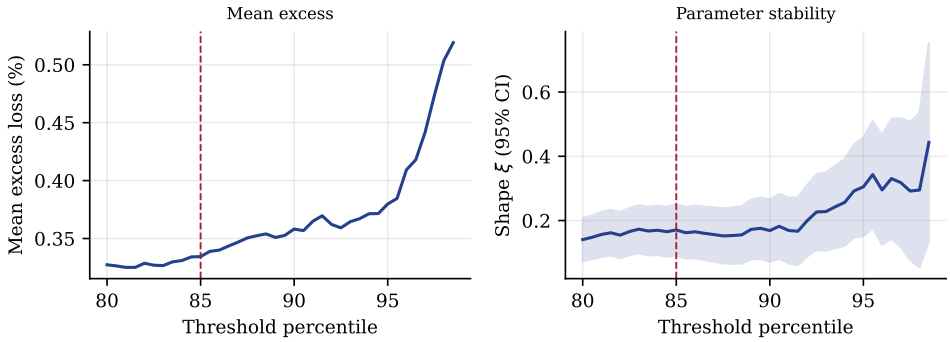


Figure 3: Threshold diagnostics. Left: mean excess loss rises with the threshold, the signature of a heavy tail ($\xi > 0$). Right: the shape estimate and its 95% interval are flat from the 80th to about the 92nd percentile. The dashed line is the chosen threshold. Source: Team analysis.

The rule selects the 85.0% percentile, a daily loss of 0.33%, leaving 828 exceedances, and a shape of $\xi = 0.171$ (standard error 0.041). A positive shape means a power law tail: large losses become rarer only slowly as they grow.

Does the choice matter? Under three common alternative rules (Table 5) the 99% ES moves by only $6,766. The 99.9% VaR moves more because it extrapolates, and the square root rule's 74 exceedances give a noisy shape of 0.446. The number we use for limits does not depend on the threshold; numbers far beyond it do.

Table 5: EVT under four threshold rules. ES 99% and VaR 99.9% in USD, one day. Source: Team analysis.

Rule	Percentile	Exceed.	ξ	ES 99%	VaR 99.9%
Parameter stability (automatic)	85.0%	828	0.171	$179,330	$248,510
90th percentile	90.0%	552	0.169	$178,839	$247,488
95th percentile	95.0%	276	0.305	$183,103	$262,961
Square-root rule (k = $\sqrt{n}$)	98.7%	74	0.446	$185,605	$263,381

5.3 What EVT gets right, and what it misses

At 99.9%, a one in a thousand day, EVT puts the loss at $248,510; the average loss on days worse than that is $324,476. History alone cannot say this, with only five or six such days. The fit is well calibrated at that level: it predicts 5.5 exceedances in our sample and we saw 5.

The worst day, seen three ways. On 2008-10-10 the portfolio lost 5.8% ($584,732). A Normal model calls that a once in 10^{36} year event; EVT calls it once in 183 years. Twenty two years of history contain it and a 5.5% loss on 2020-03-12. EVT is far better than the bell curve, but it treats every day as drawn from one distribution, while the worst days came in weeks of extreme volatility. That is the case for GARCH-EVT (McNeil and Frey, 2000), which filters volatility with a GARCH model (Bollerslev, 1986) first: its ES today is $139,652 because markets are calm, and it rises as soon as they are not.

6. Layer 5: was the risk model actually right?

6.1 Design

A risk model is only as good as its record on data it has not seen. Every forecast here uses only the 500 trading days before the forecast day, and is compared with that day's actual return. Slow models are refitted every 20 days. The test starts on 2007-01-03, the first date with 500 days of history behind it, so it includes the whole 2008 crisis, and runs to 2026-10-26: 4,985 days. We apply Kupiec (right number of breaches), Christoffersen (are breaches independent?) and the

Basel traffic light (Table 6, Figure 4).

Table 6: Out of sample backtest of 99% one day VaR, 4,985 days from 2007. Zone is the Basel traffic light for the last 250 days; "time in red" is the share of all rolling 250 day windows in the red zone. Source: Team analysis.

Method (99%, 1-day)	Breaches	Expected	Kupiec p	Indep. p	Cond. cov. p	Zone	Time in red
Historical simulation	64	50	0.054	0.000	0.000	Green	4.9%
Parametric (Normal)	99	50	0.000	0.000	0.000	Yellow	14.9%
Parametric (Student-t)	84	50	0.000	0.000	0.000	Yellow	13.3%
Cornish-Fisher	39	50	0.108	0.000	0.000	Green	0.0%
Monte Carlo (Normal)	104	50	0.000	0.000	0.000	Yellow	18.0%
Monte Carlo (Student-t)	79	50	0.000	0.000	0.000	Yellow	13.1%
Filtered HS (EWMA)	59	50	0.206	0.038	0.052	Green	0.0%
EVT (GPD-POT)	61	50	0.125	0.000	0.000	Green	9.8%
GARCH-EVT (conditional)	60	50	0.162	0.001	0.001	Green	1.6%

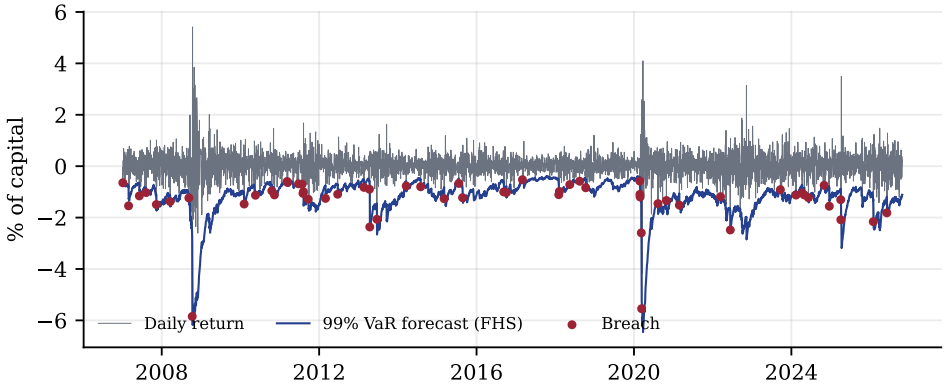


Figure 4: Out of sample 99% VaR from filtered historical simulation against actual daily returns. Breaches are marked in red. Source: Team analysis.

6.2 Results

The Normal model fails clearly: 99 breaches against 50 expected (Kupiec p = 0.000), and 14.9% of the period in the red zone. This is Section 4's fat tail problem, measured out of sample.

5 models pass on the count at 5%: historical simulation only just (p = 0.054), Cornish–Fisher by breaching too rarely, and FHS, EVT and GARCH-EVT. But EVT still spent 9.8% of the period in the red zone and historical simulation 4.9%, because their breaches came in bursts, above all in 2008 and 2020.

Only FHS passes conditional coverage, and only just. Every other model fails the Christoffersen tests outright. FHS marginally passes conditional coverage, the joint test of count and timing (p = 0.052), and never entered the red zone, including 2008. It still fails the independence test on its own (p = 0.038): its breaches cluster less than any other model's, but they do cluster, which is why trigger (c) watches for bunching. It holds up in both halves of the test: 19 breaches against 13 expected in 2007–2011 (p = 0.09) and 40 against 37 in 2012–2026 (p = 0.65).

ES and the 95% level. The Acerbi–Székely Z2 statistic of 0.26 says losses on breach days were somewhat larger than the model's ES, but inside the level at which their test rejects. At 95% the model breached 257 times against 249 expected (Kupiec p = 0.62), yet 18 of those fall in the last 250 days, so the 95% model is in the yellow zone today. Basel defines zones only at 99% (green up to 4 breaches in 250 days, red from 10); we apply the same probability cut-offs at 95%, which gives green up to 17 and red from 27. We treat this as an early warning, and it feeds trigger (c) in Section 10.

7. Stress testing: real crises and two we invented

7.1 Historical replays

We replayed six crises on the recommended weights, held without rebalancing as a real portfolio would be in a fast sell-off (Table 7): the brief's 2008, March 2020 and 2022, plus 2011, the 2013 taper tantrum and late 2018.

Table 7: Historical crisis replays, peak to trough of each market episode. "Max DD" is the deepest fall inside the window, which the stress limit is tested against. Replays are not rebalanced, so weights drift during the window; that is why the 2008 figures differ slightly from the daily rebalanced worst fall (–18.7%) and worst day (5.8%) quoted elsewhere. The 2020 and 2022 deepest falls are both –13.9% to one decimal. Source: Team analysis.

Historical scenario	Window	Ours	Max DD	Worst day	Loss (USD)	60/40
2008 Global Financial Crisis	2007-10-09 to 2009-03-09	–4.2%	–18.1%	–6.3%	$419,479	–30.0%
2011 US downgrade & euro crisis	2011-07-22 to 2011-10-03	–2.0%	–4.1%	–1.4%	$199,798	–9.3%
2013 Taper tantrum	2013-05-22 to 2013-06-24	–4.8%	–4.8%	–2.1%	$476,081	–4.8%
2018 Q4 sell-off	2018-09-20 to 2018-12-24	–3.2%	–3.7%	–0.8%	$319,973	–10.5%
March 2020 COVID crash	2020-02-19 to 2020-03-23	–10.7%	–13.9%	–5.2%	$1,066,531	–20.6%
2022 Rate-hiking drawdown	2022-01-03 to 2022-10-12	–12.8%	–13.9%	–2.6%	$1,276,405	–20.4%

2008. The portfolio ended the window down only –4.2% but fell –18.1% on the way, its deepest fall in any scenario; the 60/40 fund fell –30.3%. The end to end number flatters, so the limit is tested on the deepest fall. **March 2020.** It lost –10.7% in five weeks, half the 60/40 loss. On 2020-03-12 all seven funds fell together, gold and AGG included. **2022.** The worst end to end loss, –12.8% or $1,276,405. Stocks and bonds fell together as inflation forced rapid rate rises; gold held and XLE rose, and the portfolio lost far less than the 60/40 fund's –20.4%. **2013.** The one episode where our defensive tilt bought nothing: a pure rate shock hit bonds and gold together, and we lost the same as 60/40.

7.2 Scenarios we invented

History holds only the crises that happened. We designed two aimed at the portfolio's weak point, the assumption that bonds and gold protect equities (Table 8).

Stagflation with emerging market debt stress. Inflation re-accelerates, the Fed hikes again, a strong dollar triggers emerging market debt stress and oil spikes on a supply shock. Equities fall 22% to 35% and bonds 10%; gold and energy gain. The portfolio loses –8.6%, less than in 2022, because gold and energy are built to pay in exactly this case. It tests the oil and emerging market channel rather than the bond channel.

Stock and bond correlation breakdown. A deeper 2022: equities fall much further (SPY 30%, against 18% in calendar 2022), bonds slightly more (15%, against 13%), and gold falls too, so nothing diversifies. It costs –16.8%, or $1,675,000, against –24.0% for 60/40: our worst invented scenario, still inside the 25% limit, and shallower than the –18.1% we actually lived through in 2008.

Table 8: Invented scenarios: instant shocks to each fund and the resulting portfolio return. Source: Team analysis.

Scenario	SPY	EFA	EEM	AGG	GLD	XLE	XLK	Ours	60/40
Stagflation with EM debt stress	–22%	–25%	–35%	–10%	8%	5%	–32%	–8.6%	–17.2%
Stock-bond correlation breakdown	–30%	–32%	–38%	–15%	–5%	–20%	–40%	–16.8%	–24.0%

8. The stock and bond relationship has changed

The portfolio's defensive premise is that bonds rise when equities fall. Campbell, Pflueger and Viceira (2020) explain when that holds. When growth shocks dominate, inflation rises in booms,

bond prices fall in good times, and bonds hedge stocks. When inflation shocks dominate, inflation rises in bad times and bonds fall with stocks. They date the switch to bonds-as-hedge around 2001 with a break test on daily returns. Their sample ends in 2011, so we ran the same test on ours.

The test. We regress AGG's daily return on SPY's, allowing the slope to change at an unknown date, and take the date with the largest F-statistic (a Quandt likelihood ratio test, Andrews 1993), with returns winsorised and Newey–West standard errors. The break falls on **26 March 2020**, with F = 86 against a 5% critical value of 8.68. AGG's correlation with SPY (Figure 5) moved from −0.08 before the break to 0.20 after it, and its slope on SPY from −0.02 to 0.06. Removing mid February to April 2020 still finds a break in May 2020 (F = 93), so the crash days themselves are not the cause.

Why 2020, and what came next. The F-statistic exceeds its critical value for candidate dates from March 2008 to July 2023, so the fact of a break is more certain than its exact date. March 2020 is when the Fed cut to zero and bond prices had no room left to rise in a sell-off. Testing the period after June 2020 alone finds a second break on 29 March 2022 (F = 16), two weeks after the Fed's first hike into the inflation shock, with correlation rising from 0.05 to 0.25. That is the inflation-shock regime Campbell, Pflueger and Viceira describe.

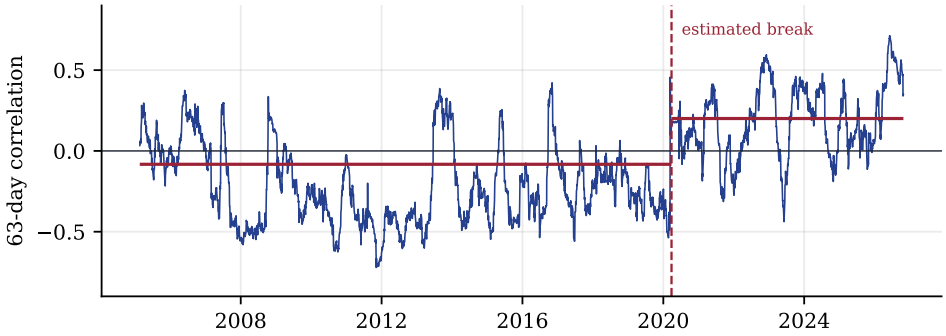


Figure 5: Rolling 63 day correlation of SPY and AGG daily returns. The dashed line is the estimated break; horizontal lines are the correlation of all days either side, the same figures as in the text. Source: Team analysis.

What it means for the client. The SPY–AGG correlation is 0.43 in 2026 so far. AGG's own volatility rose from 4.6% before 2020 to 6.3% since. Table 11 shows the limits on recent data: from 2020 the portfolio breaches the volatility and ES limits, mostly because of March 2020; from 2021 it passes all four checks that can be measured on recent data, with less headroom than before. In an inflation shock regime the holdings that pay are gold and energy. That is the lever we ask the committee to pre-approve: lifting gold from 20% to 25%, funded from AGG ("Gold 25" in Table 3). It is a small trade: it loses −12.5% in 2022 against our −12.8% and −16.3% in the correlation breakdown against −16.8%, and it uses up headroom (volatility 7.8%, ES 1.93%, worst fall −19.3%, or −19.9% rebalanced quarterly). So it is conditional: it happens only if the new mix still passes every limit on quarterly rebalanced figures at the time.

9. Where does the risk come from, and how big should each position be?

Capital and risk are not the same thing. We split the 99% ES into each fund's contribution (an Euler allocation, Tasche 1999): each fund's weight times its average return on the portfolio's worst 1% of days. The contributions (Table 9) add up exactly to the simple return ES of $181,158; Table 4's $181,583 is measured on log returns.

The equity sleeve is 25% of the capital but 57.1% of the tail risk; SPY alone, at 15% of the money, carries 31.3%. AGG is the reverse: 55% of the money and 15.0% of the risk. XLE takes the most risk per dollar, 3.06 times its weight, which is why it is capped at 5%.

The budget sets the sizes. Table 10 moves 5% of capital from AGG into each fund in turn.

Table 9: Risk budget for the recommended mix. "Risk / capital" is a fund's share of ES divided by its weight; above 1 means it takes more than its share of risk. Normal component VaR is shown for comparison. Source: Team analysis.

Fund	Weight	Share of VaR	Share of ES	Component ES (USD)	Risk / capital
SPY	15%	27.0%	31.3%	$56,765	2.09
AGG	55%	18.7%	15.0%	$27,189	0.27
GLD	20%	32.1%	27.9%	$50,587	1.40
XLE	5%	12.4%	15.3%	$27,732	3.06
XLK	5%	9.8%	10.4%	$18,884	2.08

Any further 5% of equity takes the 99% ES to between 1.99% and 2.10%: SPY and XLK land just inside the limit and XLE, EFA or EEM breach it. The equity sleeve is therefore at its budget; gold is the only addition that stays well inside. This is a boundary check on the current weights, not a full risk-parity sizing: it shows the limit, not the budget, is what stops more equity. If the conservative ES ever exceeds 2%, weight moves from the equity sleeve into AGG in proportion to each fund's ES contribution, and ES is recomputed before any trade. Cash is not in the permitted universe, so AGG is where risk is moved to.

Table 10: Position sizing: the historical 99% one day ES after moving 5% of capital between funds, starting from the recommended mix. Source: Team analysis.

Move 5% of capital	99% ES after	Change	Within 2% limit
AGG → GLD	1.93%	+0.11 pp	Yes
AGG → SPY	1.99%	+0.18 pp	Yes
AGG → XLK	2.00%	+0.18 pp	Yes
AGG → XLE	2.10%	+0.28 pp	**No**
AGG → EFA	2.01%	+0.19 pp	**No**
AGG → EEM	2.07%	+0.25 pp	**No**
GLD → AGG	1.73%	−0.08 pp	Yes

10. Turning the numbers into a decision

Table 11 sets the evidence against the limits, on the full sample and on recent data. All 9 checks pass on the full sample. The tightest are ES and volatility, because the equity sleeve was sized to the ES budget (Section 9). On 2020–26 data volatility and ES breach, driven by March 2020; on 2021–26 data all four checks that can be measured on recent data pass. We judge the mandate on the full sample, because a five year mandate will span more than one regime, and hand the recent reading to the triggers below. Read on the same quarterly basis as the trigger mix, the recommended mix returns 7.1% (0.6 points above today's target), falls −19.2% at worst and has a 99% ES of 1.77%.

Our decision rule. The brief offers three answers: accept, adjust the risk limits, or reject. The app applies them mechanically to the hard limits in Table 11 and the 99% backtest: reject if two or more fail or the backtest is red; adjust if one fails or the backtest is yellow; accept otherwise. Every hard limit passes and the 99% backtest is green, so the answer is the brief's first: **accept**. The 95% yellow zone and the correlation reading are monitoring signals that feed the triggers, not mandate checks. Setting the drawdown tolerance at 20% was part of designing the mandate (Section 3.3), not a concession made after the fact.

Why accept with two triggers live on day one. The correlation trigger and the 95% yellow zone are both on today, but neither is a breach, both point to the same risk (a weaker bond hedge), and the response to it is agreed in advance rather than improvised in a crisis.

Table 11: Mandate scorecard. ES is the largest of filtered historical simulation, EVT and historical simulation. Losses are shown as positive numbers; bold marks a breach; headroom is computed on unrounded figures. The last two columns repeat the four checks that can be measured on recent data. Source: Team analysis.

Check	Full sample	Limit	Headroom	Status	2020–26	2021–26
Excess return over T-bills	5.5%	≥ 2.5%	3.0 pp	Pass	5.5%	4.2%
Return against today's target	7.3%	≥ 6.5%	0.7 pp	Pass	–	–
Annualised volatility	7.3%	≤ 8%	0.7 pp	Pass	**8.6%**	7.7%
99% 1-day ES (conservative)	1.82%	≤ 2%	0.2 pp	Pass	**2.17%**	1.61%
Maximum drawdown, 2004–2026	18.7%	≤ 20%	1.3 pp	Pass	14.4%	14.0%
Deepest fall in a crisis replay	18.1%	≤ 25%	6.9 pp	Pass	–	–
Worst invented scenario	16.8%	≤ 25%	8.2 pp	Pass	–	–
99% VaR backtest (FHS), last 250 days	Green	Green	–	Pass	–	–
Allocation limits	Met	All met	–	Pass	–	–

Recommendation.

1. **Accept the mandate.** Invest USD 10 million as AGG 55%, GLD 20%, SPY 15%, XLE 5%, XLK 5% (USD 5.5m, 2.0m, 1.5m, 0.5m, 0.5m). Rebalance to these weights at each quarterly review.
2. **Set the drawdown tolerance at 20%,** about five years of payout, and make a 15% fall trigger a formal committee review.
3. **Hold risk against the conservative ES.** Measure the 99% one day ES daily as the largest of FHS, EVT and historical simulation, keep it below $200,000, and report the ten day and stressed figures ($563,023 and $322,842) alongside it, for information.
4. **Pre-approve three triggers and their actions.** (a) *Correlation:* if the rolling quarter's SPY–AGG correlation stays above its post-break level of 0.20, move 5% from AGG to gold, lifting the gold cap to 25% for that purpose, provided the new mix still passes every limit on quarterly rebalanced figures. Since the break the reading has been above 0.20 on 45% of days, so it will often be on; the limit test is what stops it becoming constant trading. On current readings the committee decides this at the first meeting. (b) *Conditional ES:* if the conservative ES exceeds 2%, move equity into AGG under the sizing rule in Section 9 until it is back inside. (c) *Clustered breaches:* if the 99% model enters the Basel yellow zone, or the 95% model stays yellow for two months (it is yellow today), the committee reviews the model.
5. **Re-estimate monthly, re-validate yearly.** Refit all models monthly; rerun the backtest, stress tests and break test at each annual review.

11. The questions we expect from the committee

Why not raise the AGG cap and keep 15%? Because it does not work for a portfolio with equities. With at least 15% in equities the best drawdown in our grid is –15.8% even with AGG uncapped. Meeting 15% means holding about 5% in equities and earning 5.2%, below today's 6.5% target. The real choice is between 15% and the return the scholarships need.

Why not simply hold a 60/40 fund? It earned 8.1% a year against our 7.3%, but fell –34.7% at worst and has a 33.0% chance of a 20% fall in five years (ours 3.5%).

Why not the constrained Markowitz mix, which scores slightly better? Its edge, 1.09 against 1.07, comes from full sample average returns, and Section 3.4 shows such picks lose most of their edge out of sample. It also lost –16.8% in 2022, against our –12.8%, in the regime Section 8 says we are now in.

The trigger is expected to fire, so why not start at 25% gold? Over this sample it does score

better: 7.7% a year against 7.3%, a higher Sortino, and smaller losses in 2008 and 2022 (the same in 2020). We pay for that in headroom and in dependence on gold: volatility 7.8% against an 8% limit, ES 1.93% against 2%, and a quarterly rebalanced worst fall of −19.9% against 20%, with gold already supplying 66% of our alpha. A mix that sits at three limits at once is for the committee to choose with current data, not for us to choose in advance.

Why hold 20% in gold, which pays no income? Because it protects in the crisis bonds do not: in 2022 AGG lost −13.0% and gold −0.8%. Its strong run supplied 66% of our alpha, which is why the cap rises only by pre-agreement and only when the correlation trigger fires.

Can we trust the risk model? It has the right number of breaches over 4,985 unseen days, including 2008, and is the only model of nine that passes conditional coverage, though marginally and with some clustering left. The usual default, the Normal model, breached 99 times against 50.

What would make us change it? A red 99% backtest, ES above 2% after the sizing rule, or a 15% fall. Each is measured by the app, not left to judgement.

12. Assumptions, limitations and wider risks

(i) **One historical path, seen in advance.** We searched 1,601 mixes before choosing, so the full sample ratios of our mix are selected figures. The walk forward and sub period tests limit this bias; they cannot remove it.

(ii) **The return record leans on gold and falling rates.** 66% of the alpha comes from gold, and bonds benefited from two decades of falling yields. Against today's 6.5% target the historical 7.3% leaves only 0.7 percentage points, and the bootstrap puts the chance of missing the target over five years at 17.7%.

(iii) **Model simplifications.** Risk models assume daily rebalancing while replays hold the starting weights; ten day figures use the square root of time; EVT beyond 99.9% is an extrapolation (shape standard error 0.041); the break date is uncertain; and with 50 expected breaches the backtests have limited power.

(iv) **The invented scenarios are judgements.** The stagflation scenario costs less than 2022 because gold and energy rally in it; the correlation breakdown is the harder test.

(v) **US concentration.** All seven funds are US listed and four of our five holdings are US assets, so the portfolio is exposed to a US specific shock.

(vi) **Currency and inflation.** Returns are nominal US dollars. If the endowment spends in Sri Lankan rupees, USD/LKR moves add risk none of these numbers contain, and the committee should decide separately how much to hedge.

(vii) **Costs, liquidity and spending.** Trading costs, taxes and panic price gaps are not modelled. The 4% spending rule is our assumption, and a fixed pool is assumed: paying out during a downturn locks in losses, so drawdowns net of payouts would be deeper than the −18.7% shown.

The app's Assumptions & Limitations page states each of these in plain language, and lets the committee change any limit and see the effect.

Appendix A. Use of AI tools

An AI assistant (Claude, by Anthropic) was the only AI tool used. It helped us read the brief, proposed the first draft portfolio (tested and rejected, Section 1), helped write and debug the code, and drafted this memo's prose from our results. We checked its output in four ways. The 24 tests in `tests/` compare the maths with answers known in advance (`python -m pytest`). Every number, table and figure here can be reproduced in our app from the same frozen data. Every claim the assistant made about the funds was tested against the data. Each module has an owner on the team who reviewed it. The mandate limits, the final portfolio, the invented scenarios, the interpretation and the recommendation are our own decisions, and every member can explain them.

Appendix B. The application and reproducibility

The Streamlit app runs with `streamlit run app.py`. Its sidebar sets the confidence level (90, 95 or 99%), VaR method and backtest dates; its pages cover the recommendation and alternatives, portfolio, a VaR/ES/EVT dashboard, EVT diagnostics, backtesting with the traffic light, stress tests, the risk budget, data diagnostics and the Assumptions & Limitations page.

Appendix C. Model settings

Model	Setting
Covariance	Sample covariance of daily simple returns, annualised by 252; Ledoit and Wolf (2004) shrinkage is available in the app as a robustness check.
EWMA (for FHS)	RiskMetrics decay $\lambda = 0.94$, one step ahead, started from the variance of the first 30 days.
GARCH(1,1)	Constant mean, Gaussian quasi maximum likelihood, $\alpha + \beta < 1$; refitted every 60 days in the backtest with the variance updated daily.
Monte Carlo	50,000 draws of all seven funds from a multivariate Normal, or a multivariate Student-t with the portfolio's fitted degrees of freedom (clipped to 3 to 30), both with the sample mean and covariance via a Cholesky factor.
EVT	Generalised Pareto by maximum likelihood; thresholds tested from the 80th to 98.5th percentile in 0.5 point steps.
Bootstraps	ES interval: 1,000 independent resamples. Five year paths: 3,000 paths of 21 day blocks.
Sortino	Downside deviation below the T-bill rate.
Backtest	500 day rolling window, refit every 20 days, first forecast 2007-01-03.
Break test	15% trimming, returns winsorised at 0.5% and 99.5%, Newey–West with one lag.

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